

LIYU XIAO

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Education

McMaster University

(B.A.Sc.) in Computer Science

Expected May 2028

Hamilton, ON

- Relevant Coursework: Data Structures & Algorithms, Logic, Software Development, Distributed Databases

Experience

Bot Auto

Software Engineer

Sept 2026 – Dec 2026

Houston, TX

- Incoming F26 (4 month contract)

Wealthsimple

Software Engineering Intern

Jan 2026 – Aug 2026

Toronto, Canada

- Developed a financial hold management system in Kotlin, implementing a GraphQL mutation and DAO layer to allow modification of check deposit holds, mitigating an estimated **\$150K/month** in fraudulent checks.
- Offloaded logging from the critical transaction path to an Avro/Kafka to S3 to Snowflake pipeline in Kotlin, Airflow, and Terraform, cutting order reservation latency by **80%** while ingesting **5 million** records/day.
- Decommissioned the legacy Postgres write path, reducing service table storage by **98.33%** from **30TB** to **500GB**
- Engineered a full-stack document generation engine across 6 microservices (Ruby, Python, React) using GraphQL and Jinja templates, enabling **8M+** beneficiaries to access RESP contribution and grant data on Wealthsimple's platform
- Architected a Ruby-based financial calculator service for QESI grants across 4 phased releases, extending the core GraphQL schema to provide **1M+** Quebec-based clients with granular repayment and withdrawal breakdowns.
- Deployed an RBAC system and production DB migration, removing cross-team dependencies for document generation.

IBM

Software Engineering Intern

Sept 2025 – Dec 2025

Toronto, Canada

- Built IBM's new *Activities* content type end-to-end, enabling **25,000+** authors across **100,000+** instances.
- Designed the Rails/PostgreSQL backend (versioning, reviews, OLX export, views), cutting DB cluster calls by **66%**.
- Added database join and query support for *Activities* across **10,000+** total assets and **20,000+** asset associations.
- Backed up and migrated platform databases, successfully updating **35,000+** records without data loss or downtime.
- Developed advanced filtering and categorization tools improving discovery of **9,000+** environmental labs.

Qoherent

Software Engineering Intern

May 2025 – Aug 2025

Toronto, Canada

- Processed **2,700** recordings into an HDF5 dataset via a Python pipeline, reducing generation time by **99.33%**.
- Trained a PyTorch model to classify 4 modulation types using **21,600** data samples, achieving **85%** accuracy.
- Deployed a customizable Docker image workflow with pre-installed tooling and support for user-defined scripts, enabling installation of radio-focused libraries and software; reduced CI job startup time by **60%**.
- Optimized the upload flow to support file selection and extension-based tracking, reducing upload time by **99%**.

Projects

Cold Email Agent | Python, Celery, Redis, FastAPI, PostgreSQL, Claude API, Firecrawl

- Architected an autonomous cold email pipeline orchestrating **4** Celery workers, processing **200+** leads/week.
- Engineered a Firecrawl-based discovery scraper covering **1,270+** startups
- Built crash-safe recovery via Celery **3**-attempt autoretry and an idempotent3 Postgres state machine
- Built a Claude tool-use extraction layer with prompt caching across **1,000+** lead calls, cutting input-token costs by **90%**

Polymarket Insider Tracker | Python, JavaScript, Web3.js, PostgreSQL

- Architected a monitoring tool using Web3.js and Python to track "smart money" wallet activity via blockchain data.
- Engineered a high-throughput backend scraper using Python to identify trade patterns with a **85%** market correlation.
- Optimized PostgreSQL ingestion pipelines to process **\$100M+** in trade volume across **50+** unique prediction markets.
- Isolated **100+** high-conviction accounts by developing a custom scoring algorithm to quantify insider trading signals.

Technical Skills

Languages & Frameworks: Python, Java, Kotlin, Ruby, TypeScript, React.js, Ruby on Rails, FastAPI

Databases: Apache Kafka, Apache Avro, PostgreSQL, Snowflake, S3, Airflow, GraphQL

DevOps & Tools: Docker, Kubernetes, Terraform, Git, Linux, CI/CD